

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11.	(a)	(i)		+3		1		1		
		(ii)		$2\text{FeAsS} + 5\text{O}_2 \rightarrow \text{As}_2\text{O}_3 + \text{Fe}_2\text{O}_3 + 2\text{SO}_2$ formulae (1) balancing (1) – only awarded if all formulae are correct award (1) for correctly balanced equation including FeO instead of Fe_2O_3 $2\text{FeAsS} + 4\frac{1}{2}\text{O}_2 \rightarrow \text{As}_2\text{O}_3 + 2\text{FeO} + 2\text{SO}_2$		2		2	1	
	(b)	(i)		moles As_2O_3 in $100\text{ cm}^3 = \frac{2.06}{197.8} = 0.0104$ (1) moles As_2O_3 in $1\text{ dm}^3 = 0.0104 \times 10 = 0.104$ (1) concentration $\text{H}_3\text{AsO}_3 = 0.104 \times 2 = 0.208\text{ mol dm}^{-3}$ (1) accept alternative method		2	1	3	2	
		(ii)		$[\text{H}^+] = 10^{-5.11} = 7.76 \times 10^{-6}\text{ mol dm}^{-3}$		1		1	1	
	(c)			pyramidal (1) contains three bonding pairs and one lone pair of electrons (1) <u>electron pairs</u> arrange themselves around the central atom as far as possible from each other so that the <u>repulsion between them is at a minimum</u> / lp – bp repulsion > bp – bp repulsion (1) first two marks can be obtained from suitable diagram		1	2	3		

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	(d)			$n = \frac{pV}{RT}$ (1) $n = \frac{1.01 \times 10^5 \times 39 \times 10^{-6}}{8.31 \times 360}$ (1) $n = 1.317 \times 10^{-3} \text{ mol}$ (1) $M_r = \frac{0.181}{1.317 \times 10^{-3}} = 137.4$ $M_r \text{ PCl}_3 = 137.5$ therefore chloride is PCl_3 (1) accept alternative method e.g. assume PCl_3 and work back to show volume of 39 cm^3	1	1				
	(e)	(i)		$(\text{P}^{35}\text{Cl}_2)^+$ accept if charge missing			1	1		
		(ii)		chlorine has isotopes ^{35}Cl and ^{37}Cl in ratio of 3:1 / 75% to 25% (1) since peak C due to $(\text{P}^{35}\text{Cl}_2)^+$ and peak E due to $(\text{P}^{37}\text{Cl}_2)^+$ height of C : E is $(3 : 1) \times (3 : 1)$ i.e 9 : 1 (1)		1	1	2		
				Question 11 total	1	10	6	17	7	0